Change Detection

IRS Chapter 15

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Quiz references: **Change Detection: Interpreting Bitemporal and Multitemporal Evidence** (Week 10).

Introduction to Change Detection in Remote Sensing

- **Change detection**: Identifies changes in surface features over time.
- Key applications:
  - Land use change
  - Vegetation monitoring
  - Environmental management

Bitemporal vs. Multitemporal Change Detection

- **Bitemporal Detection**:
  - Compares images from two specific dates.
  - Suitable for detecting abrupt changes.
- **Multitemporal Detection**:
  - Analyzes multiple dates for trend analysis.
  - Ideal for gradual or seasonal changes.
- Increasingly accessible data enables broader applications.

Conditions for Effective Change Detection

- **Consistent Sensor Data**: Same or calibrated sensors.
- **Seasonal Consistency**: Images taken in similar seasons.
- **High Co-registration Accuracy**: Preferably within 0.2 pixels.
  - This accuracy threshold comes from studies (e.g., Dai and Khorram, 1999)
- **Cloud-Free**: Clear skies over the area of interest.
- **Reflectance Correction**: Corrected to surface reflectance.

Preprocessing Steps for Change Detection

- **Image Calibration**: Ensure radiometric consistency.
- **Co-registration**: Align images to sub-pixel accuracy.
- **Reflectance Correction**: Adjust to surface reflectance.
- **Cloud Removal**: Clear cloud cover from images.
- **Noise Reduction**: Minimize environmental variation.

Visual Interpretation in Change Detection

- **Multidate Color Composite**: Combines bands from two dates.
- **Image Swiping/Flickering**: Overlay images for change visualization.
- **Side-by-Side Comparison**: View images in parallel for analysis.
- **Heads-Up Digitizing**: Manually identify and classify changes.

Image Swiping/Flickering Example

Annual NCLD Land Conver Data (from USGS) -- Oxford Area Land Cover from 2014

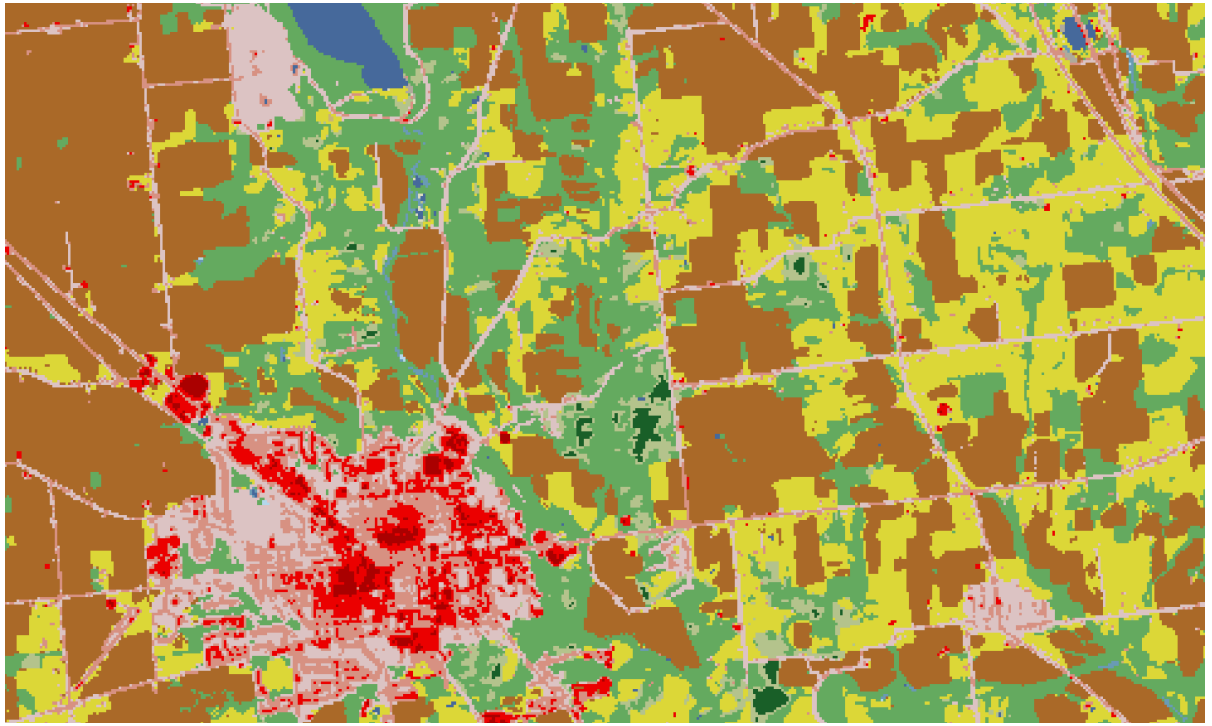


Image Swiping/Flickering Example

Annual NCLD Land Conver Data (from USGS) -- Oxford Area Land Cover from 2023

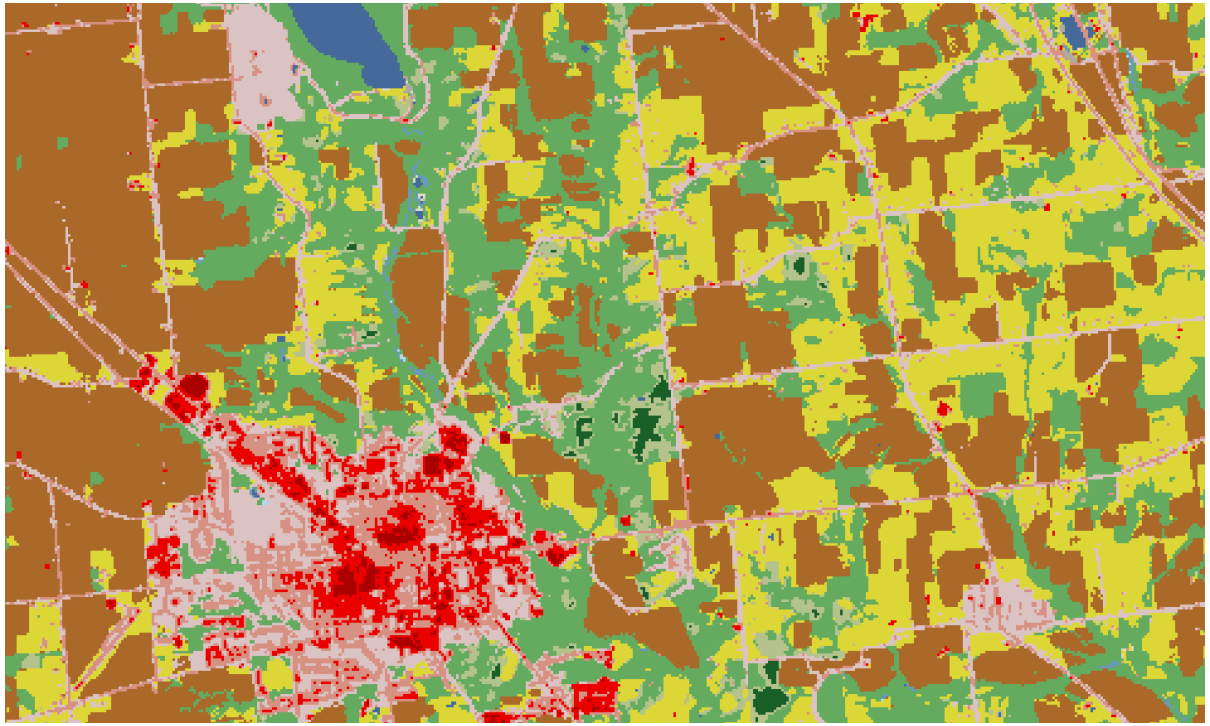


Image Swiping/Flickering Example

Aside: As programmers we can and should be able to do better

- Use `ffmpeg` to make videos or animated GIFs
- Save images as `feame-0001.png`, `frame-002.png` and so on.

```
ffmpeg -framerate 2 -i frame-%04d.png -loop 0 output.gif
```

Linear Algebra Notation and Inner Product

- **Vectors** in **bold**: e.g., $\mathbf{x}$ represents a vector.
- **Inner Product**: Measures similarity of vectors $\mathbf{x}$ and $\mathbf{y}$.

$$\mathbf{x}^T\mathbf{y} = \sum_{i=1}^{n} x_i y_i$$

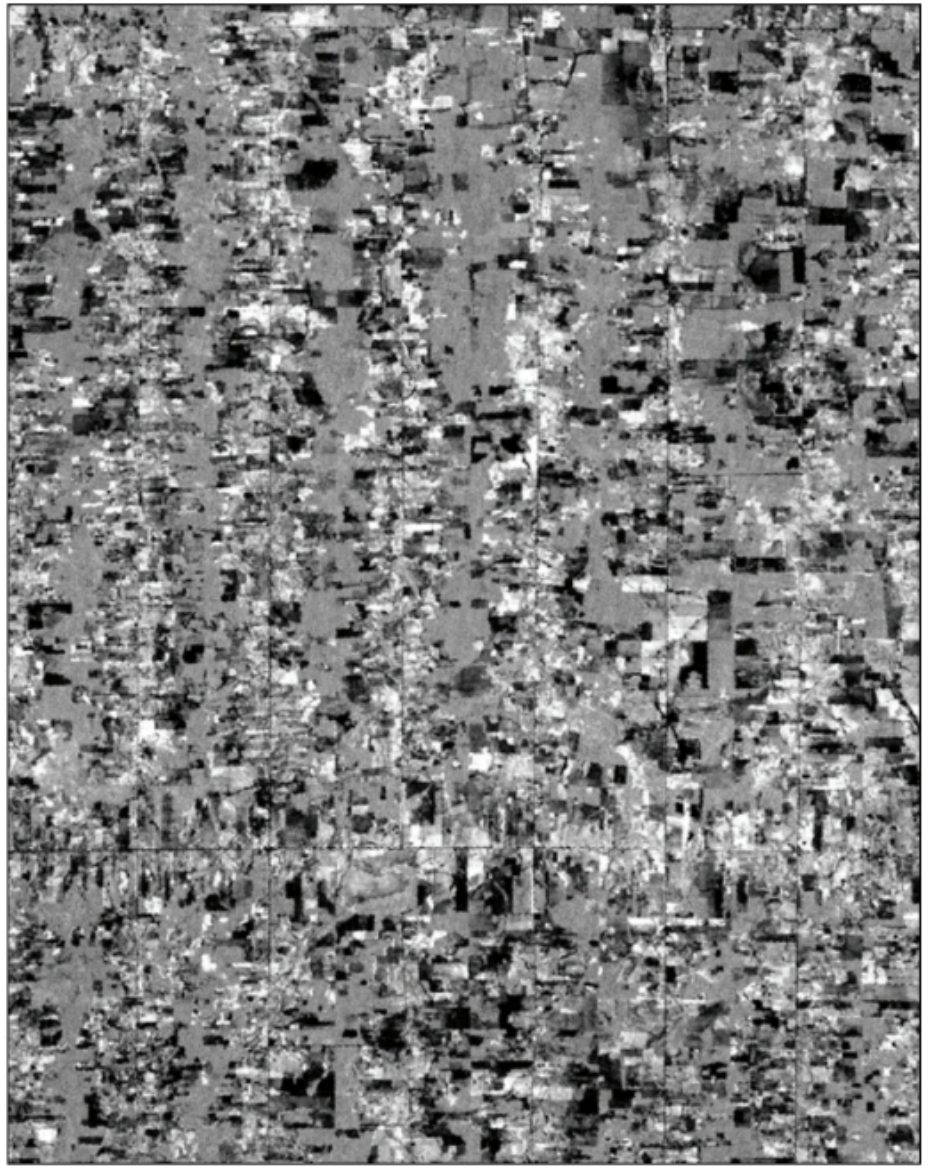
- Other names for this: ***Scaler Product***, ***Dot Product***
- Other notation $\mathbf{x} \cdot \mathbf{y}$, $\langle \mathbf{x}, \mathbf{y} \rangle$
- **Geometric Interpretation**:
  - $\mathbf{x}^T\mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$
  - Measures **projection** of $\mathbf{x}$ onto $\mathbf{y}$.
  - $\cos\theta$ gives the angle between vectors.

Image Algebra for Change Detection

- **Image Differencing**: Subtract pixel values between dates.
- **Image Ratioing**: Divide pixel values across dates. (Less sensitive to lighting & atmosphere)
- **Euclidean Distance**: Measures spectral distance.

$$D = \|\mathbf{x}_1 - \mathbf{x}_2\|_2 = \sqrt{(\mathbf{x}_1 - \mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2)}$$

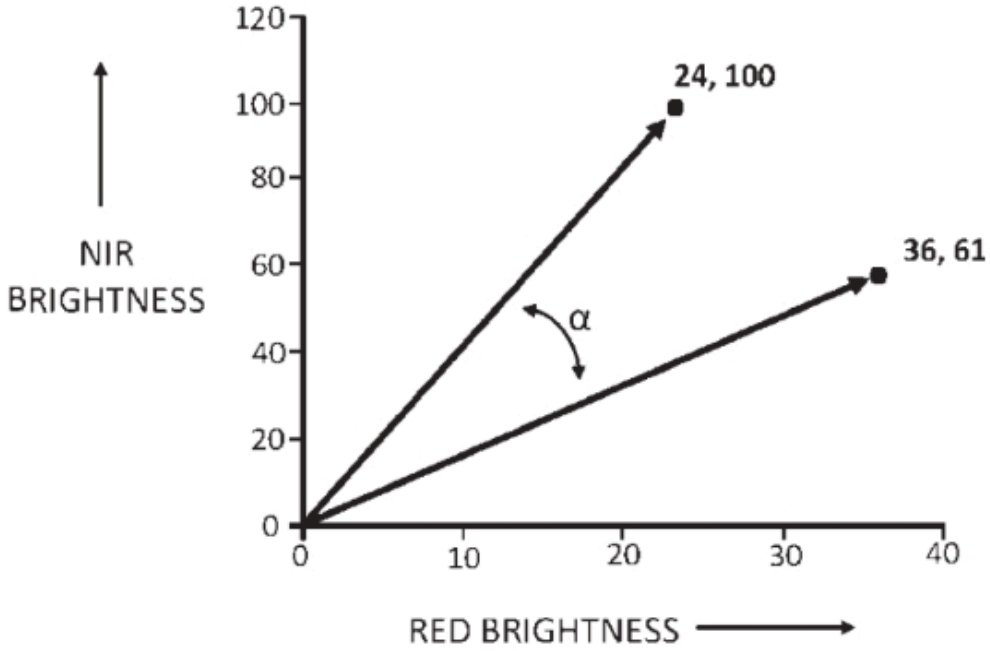
> The figure shows NDVI difference (brighter(+) = increase in vegetation, darker(-)= decrease, gray=no change)



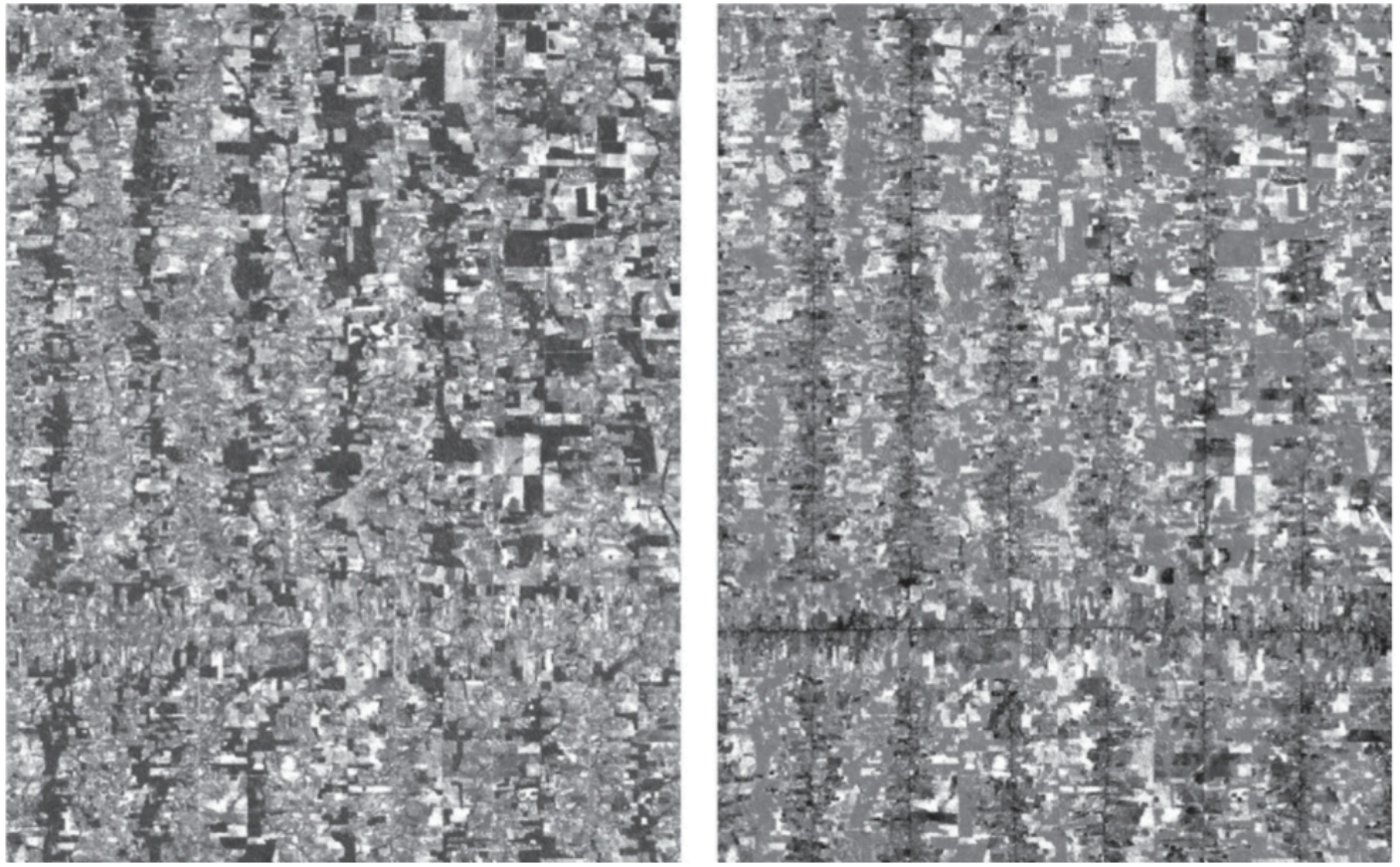
Change Vector Analysis (CVA)

- **Magnitude**: Size of the multispectral change vector.
- **Direction**: Which bands increased or decreased; useful for interpreting change.
- Spectral angle is a separate similarity measure, not CVA's change-vector magnitude.

$$\Delta \mathbf{x} = \mathbf{x}_{t_2} - \mathbf{x}_{t_1}, \qquad \|\Delta \mathbf{x}\|_2 = \sqrt{\sum_{b=1}^{B} (x_{t_2,b} - x_{t_1,b})^2}$$



Change Vector Analysis (CVA)



(LEFT) Euclidean distance, (RIGHT) Spectral Angle Measurment

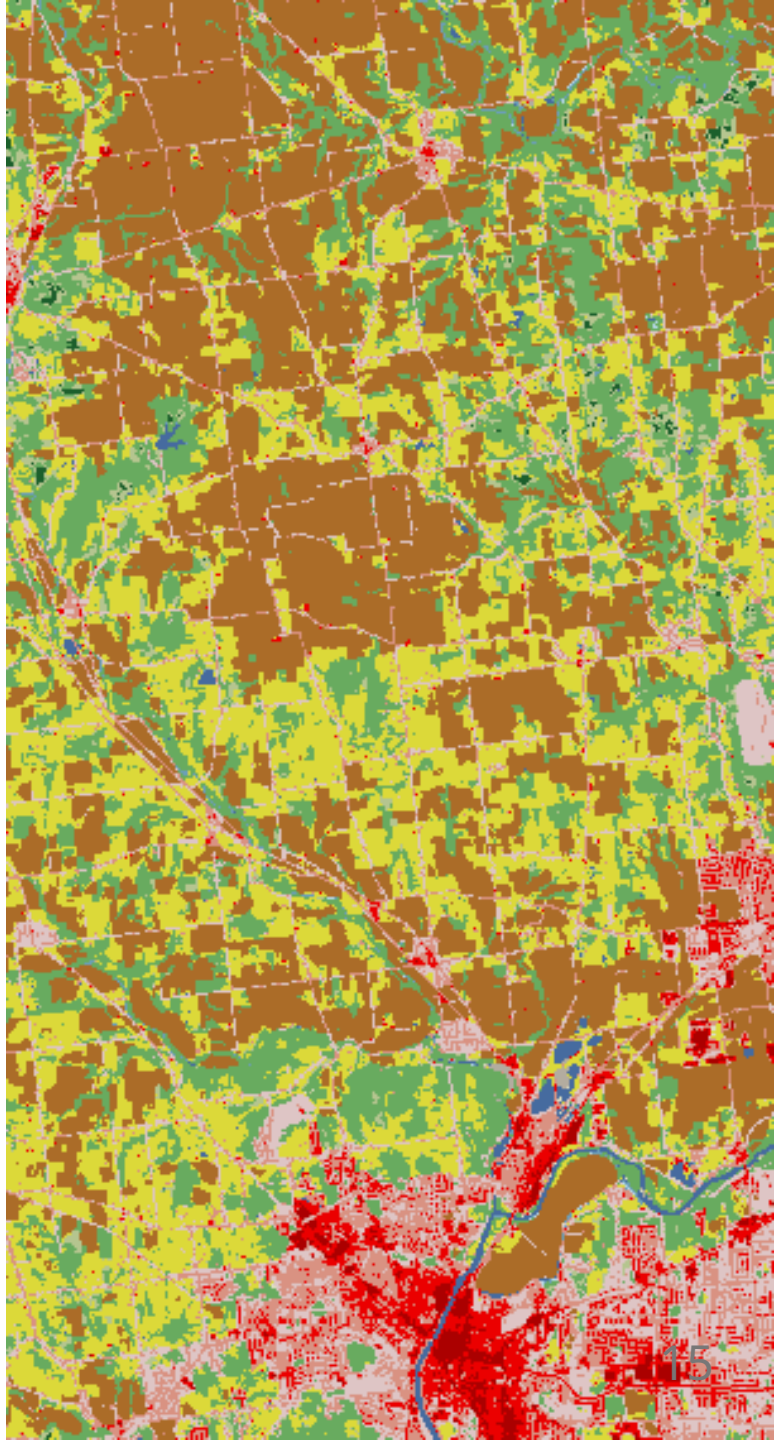
Why didn't the book include the the source images? 🤦

Transformation and Data Reduction

- **Principal Component Analysis (PCA)**: Reduces data to key components.
- **Tasseled Cap (TC) Transformation**: Analyzes brightness, greenness, wetness.
- Enables detection of prominent changes.
- These transformations simplify change detection by focusing on components that emphasize key landscape changes.

Classification in Change Detection

- **Powerful for Change Attribution**: Identifies type of land cover change.
- **Whole Area vs. Targeted Classification**:
  - Whole area: Classify all then look for changes
  - Targeted: First identify changed areas then classify the change
- **When to use each?**:
  - If whole area classification can be done accurately, this is often preferred



Correlation Is Not a Change Map

- Focus on a single band for interpretability (for now)
  - Can reduce multiband images by e.g. NDVI or PCA
- **Pearson correlation** summarizes linear association between paired values.
- A high correlation does not imply no change: adding a constant to every pixel leaves correlation unchanged.
- Use a spatial difference map to locate change; correlation alone discards location.

$$\mathrm{cov}_{12} = \frac{\sum(x_1 - \bar{x}_1)(x_2 - \bar{x}_2)}{n-1}$$

$$r = \frac{\mathrm{cov}_{12}}{s_1 s_2}$$

- $\bar{x}_1$, $\bar{x}_2$ -- are the mean instensities for image 1 and image 2
- s_1, s_2 are the standard deviations. *Why not σ?* 🤦 ¯_(ツ)_/¯

Multitemporal Change Detection: Benefits

- **Enhanced Change Detection**: Detects low-magnitude trends.
- **Gradual Change Capture**: Monitors slow processes (e.g., vegetation growth).
- **Aligned with Socioeconomic Cycles**: Annual analysis supports policy planning.

Multitemporal Change Detection: Challenges

- **Cloud Cover**: Persistent clouds complicate analysis.
- **Data Volume**: Large datasets require robust processing.
- **Phenological Effects**: Seasonal changes can obscure results.

Preprocessing for Multitemporal Detection

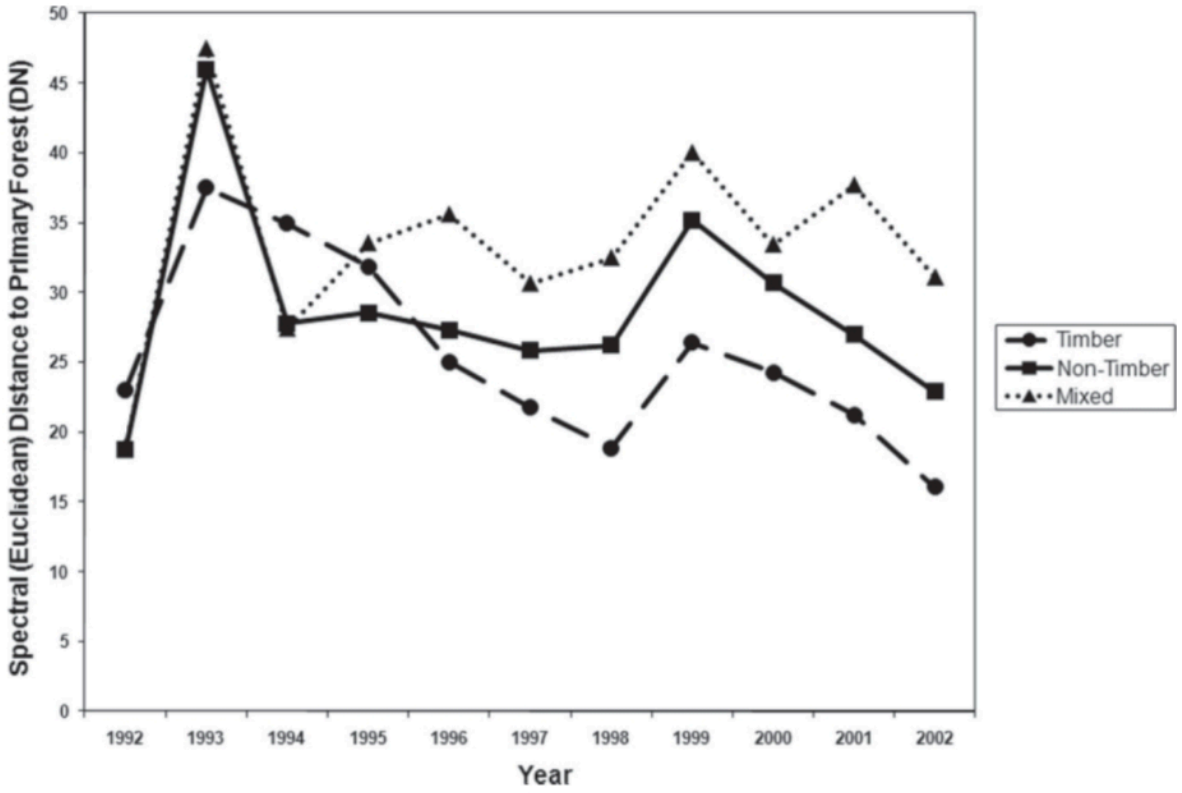
- **Take Care Collecting Images**: Aim for the same season, time of day. Register.
- **Image Co-Registration**: Also include e.g. histogram matching etc.
- **Cloud Removal**: Manual and automated techniques.
- **Image Normalization**: Adjusts for atmospheric or seasonal variance.
- **Segmentation (optional)**: Groups pixels with similar trajectories.
- **Band Selection (optional)**: Focus on a single band or derived value to simplify analysis.

Cloud Cover Analysis in Multitemporal Change Detection

- **Challenge**: Cloud cover limits clear imagery, especially with one image per year.
- **Image Selection**: Choose the clearest annual image available.
- **Gap Handling Options**: (To fill the hole left when clouds are removed)
  - **Pixel Interpolation**: Use nearby pixels in the same image.
  - **Date Interpolation**: Use nearby dates' images for missing areas.
  - Keep missing observations explicit; interpolation can hide abrupt changes.

EWMA is a monitoring statistic, not a cloud-gap interpolation method.

Example from the book....



Mean spectral distance from primary forest by agroforestry group

Time Series Analysis Basics

- **Kernel Regression**: Linear combination of basis functions.
- **Top-Down Methods**: Whole-series approximation, then partition.
- **Bottom-Up Methods**: Builds larger structures from smaller ones.

Least Squares (LS) Solution for Basis Functions

- **Goal**: Minimize error between observed values $ y $ and model predictions $\hat{y}$.
- **Model Representation**: Express $\hat{y}_x$ as a sum of basis functions:

$$\hat{y}_x = a_0 f_0(x) + a_1 f_1(x) + \cdots + a_n f_n(x)$$

- **Objective**: Find coefficients a_i that minimize the sum of squared residuals.

Example: Monomial Basis Functions in LS Solution

- **Monomial Basis Functions**: Use powers of x: $1, x, x^2, x^3$.
  - $f_i(x) = x^i$
- **Observations:** $\{(x_i, y_i) \mid \quad i = 1 \ldots m\}$
  - Note I use x rather thant t, whatever!
- **Three Cases**:
  - **Underdetermined**: Fewer points than functions (insufficient data).
  - **Exactly Determined**: Same number of points as functions.
  - **Overdetermined**: More points than functions (more data than needed).

Setting Up the Matrix for LS Solution

- **Matrix Setup**: Each element (i, j) is $f_j(x_i)$.

$$\mathbf{A} = \begin{bmatrix} f_0(x_1) & f_1(x_1) & \dots & f_n(x_1) \\ f_0(x_2) & f_1(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_0(x_m) & f_1(x_m) & \dots & f_n(x_m) \end{bmatrix}$$

- **Objective**: Solve $\mathbf{Aa} = \mathbf{y}$ for coefficients $\mathbf{a}$.

Solving with the Normal Equations

- **Normal Equations**: Used when the system is overdetermined.

$$\mathbf{A}^T\mathbf{A}\mathbf{a} = \mathbf{A}^T\mathbf{y}$$

- **Interpretation**: Projects $\mathbf{y}$ onto the space spanned by the basis functions.
- **Underdetermined Systems**: Require additional constraints or regularization.

$$(\mathbf{A}^T\mathbf{A} + \lambda\mathbf{I})\mathbf{a} = \mathbf{A}^T\mathbf{y}$$

Harmonic Basis Functions

- **Cosine and Sine Functions** as Basis:

$$\hat{\rho}_x = a_0 + a_1 \cos\left(\frac{2\pi}{T}x\right) + a_2 \sin\left(\frac{2\pi}{T}x\right)$$

- **Periodicity**: Fits seasonal changes in time series data.
- **Coefficients** a_0, a_1, a_2: Control the overall value and seasonal oscillation.
- T **(Period)**: Defines the cycle length, e.g., $T = 365$ for annual data.

Solving for Harmonic Coefficients

- **Objective**: Minimize the sum of squared residuals between observed data y_x and model predictions $\hat{\rho}_x$.
- **Set Up**: Rewrite model as a matrix equation.

$$\begin{bmatrix} 1 & \cos\left(\frac{2\pi}{T}x_1\right) & \sin\left(\frac{2\pi}{T}x_1\right) \\ 1 & \cos\left(\frac{2\pi}{T}x_2\right) & \sin\left(\frac{2\pi}{T}x_2\right) \\ \vdots & \vdots & \vdots \\ 1 & \cos\left(\frac{2\pi}{T}x_n\right) & \sin\left(\frac{2\pi}{T}x_n\right) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

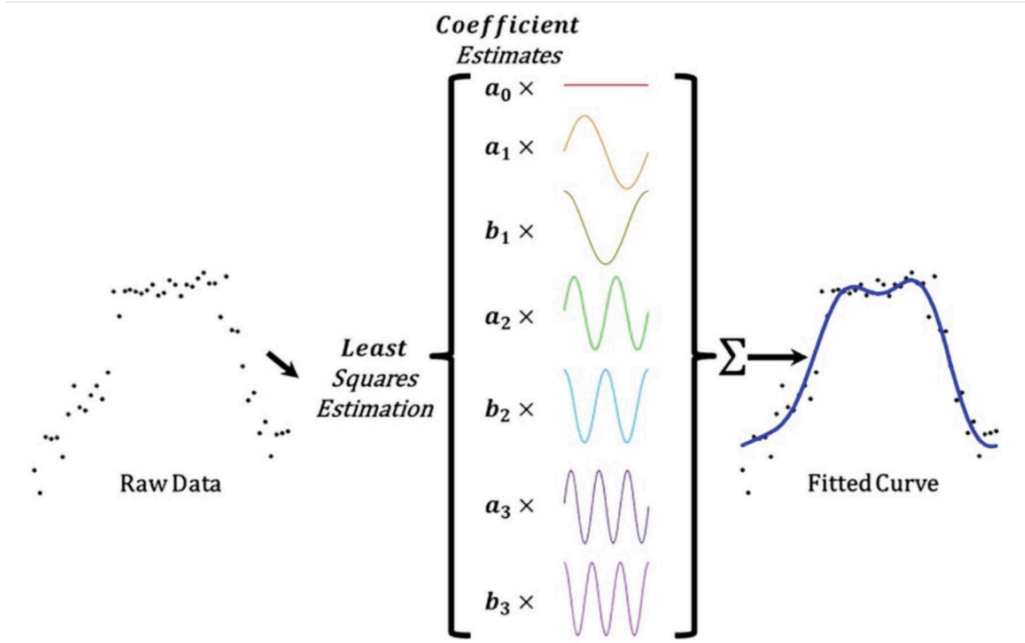
- **Solution**: Use normal equations: $\mathbf{A}^T\mathbf{A}\mathbf{a} = \mathbf{A}^T\mathbf{y}$.

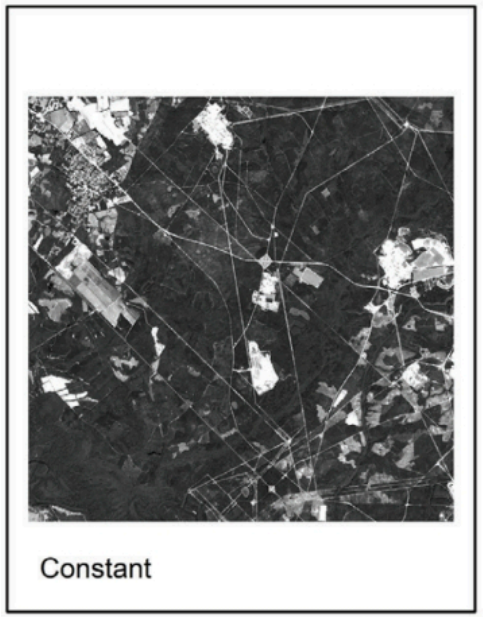
Harmonic Regression for Change Detection

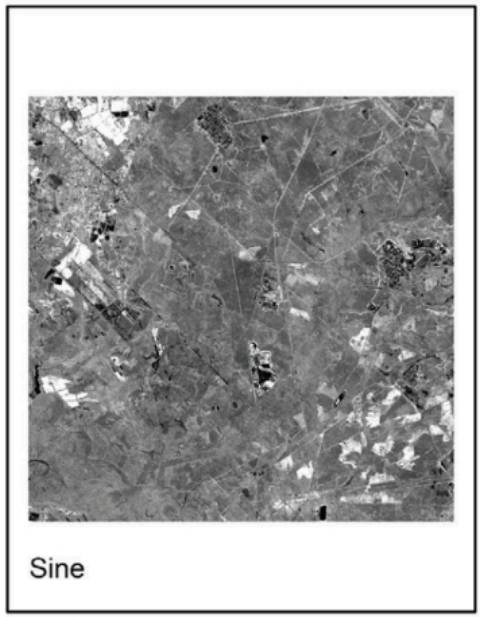
- **Models Seasonal Trends:**

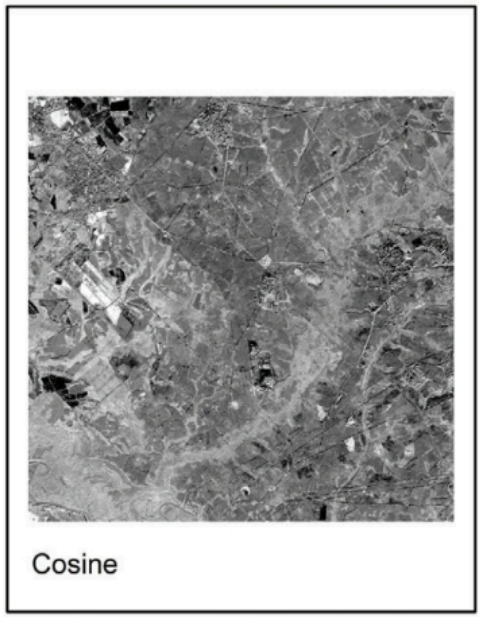
$$\hat{\rho}_x = a_0 + a_1 \cos\left(\frac{2\pi}{T}x\right) + a_2 \sin\left(\frac{2\pi}{T}x\right)$$

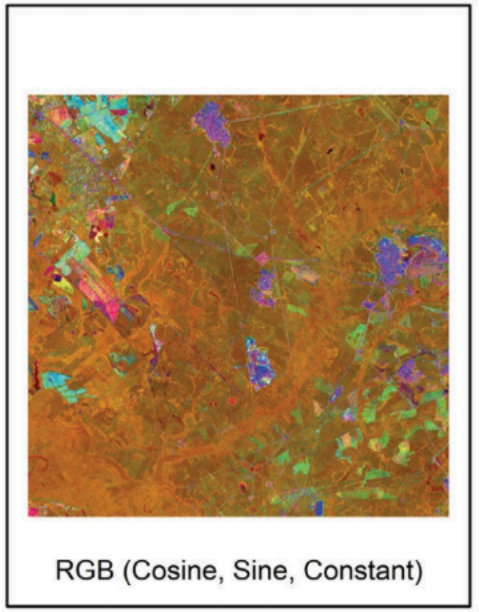
- Note that you can add new bases for different periods (months, seasons, weeks, etc)



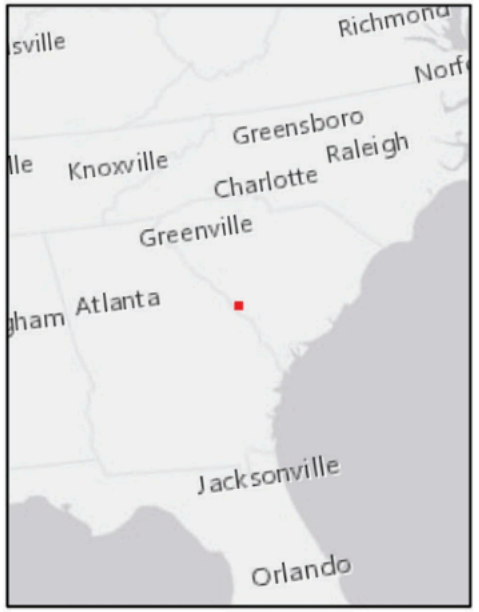

Constant


Sine


Cosine


RGB (Cosine, Sine, Constant)




Richmond
Greensboro
Raleigh
Knoxville
Charlotte
Greenville
Atlanta
Jacksonville
Orlando

Harmonic Basis and Fourier Transform

- **Fourier Series**: Generalizes harmonic regression by capturing multiple frequencies.

$$\hat{\rho}_x = a_0 + \sum_{k=1}^{N} \left(a_k \cos\left(\frac{2\pi k}{T} x\right) + b_k \sin\left(\frac{2\pi k}{T} x\right) \right)$$

- **Fourier Transform**: Decomposes signals into sine and cosine functions.
- **Fast Fourier Transform (FFT)** computes the discrete Fourier transform efficiently in $O(N \log N)$; it does not solve general least-squares regression coefficients.
- **Relation to Harmonic Regression**:
  - Harmonic regression approximates with a single frequency.
  - Fourier transform reveals all significant frequencies in data.

Control Charts in Change Detection

- **Purpose**: Monitor changes in time series data, such as vegetation or land cover.
- **Types of Control Charts**:
  - **EWMA (Exponentially Weighted Moving Average)**: Detects small, gradual shifts.
  - **Shewhart X-bar Chart**: Detects large, abrupt shifts.

The next few slides go into more detail.

EWMA Control Charts with Tolerance Bands

- **EWMA**: Exponentially Weighted Moving Average
  - **Purpose**: Smooths data to reduce noise, revealing underlying trends.
  - **Formula**:

$$\text{EWMA}_t = \lambda y_t + (1 - \lambda)\text{EWMA}_{t-1}$$

- First fit seasonality, calculate residuals, and monitor those residuals.
- **Control Limits (Tolerance Bands)**:
  - Define acceptable variation range.
  - Persistent departures can signal small or gradual shifts as well as abrupt disturbances.

Applications

- **Anomaly Detection**: Highlights abrupt shifts, such as unexpected environmental changes.
- **Preprocessing for Trend Analysis**: Filters noise to stabilize data for further analysis.

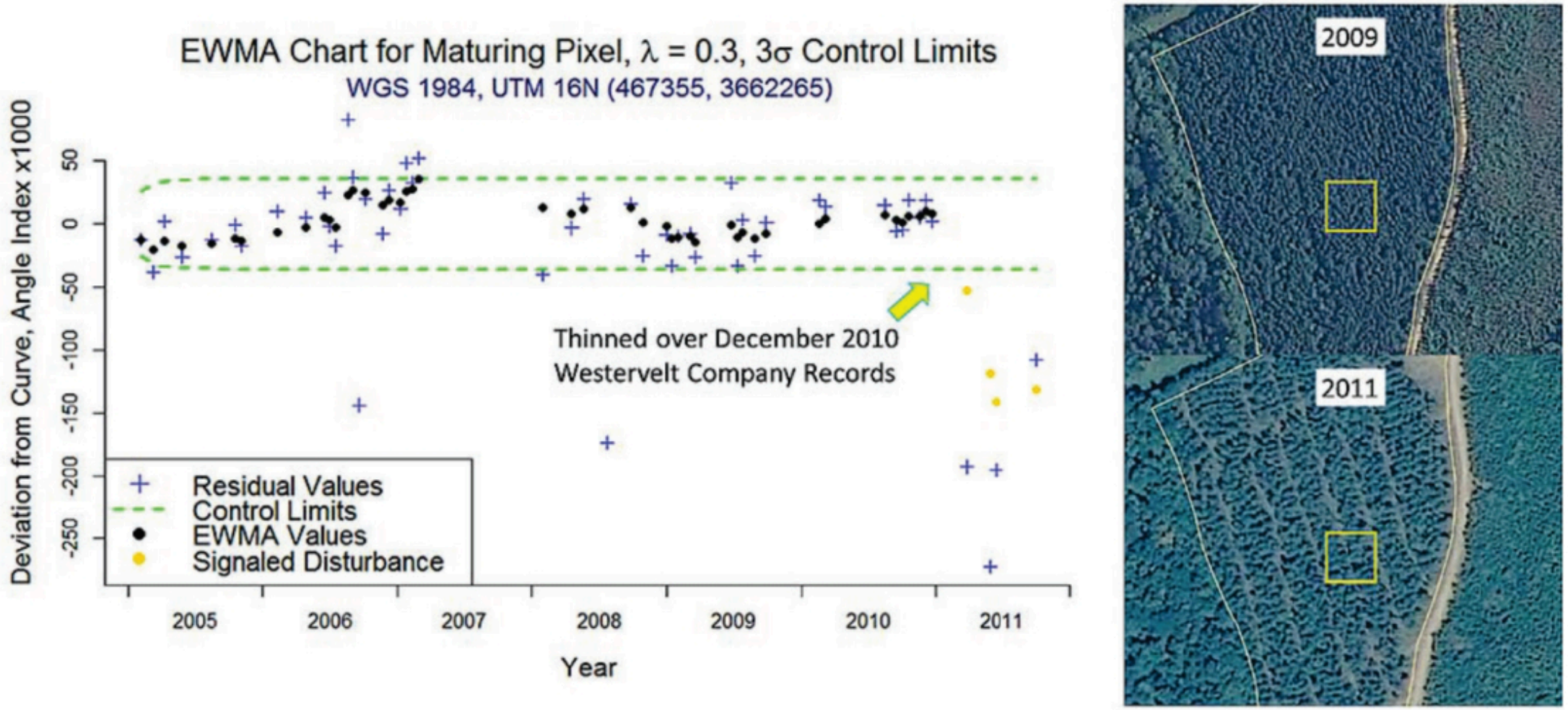


FIGURE 15.7 Exponentially weighted moving average chart for residual values after removing seasonality. The forested Landsat pixel (yellow boxes on 2009 and 2011 National Agriculture Imagery Program images at right) was thinned in December 2010.

Notebook: Compare Two Change Maps

Run the Landsat change detection demo in Colab

View the notebook as slides

Compare a robust NIR-difference threshold with a top-5% CVA ranking. Why must the latter select pixels even when no real land-cover change occurred?

Summary of Change Detection in Remote Sensing

- Change detection supports environmental and urban planning.
- **Bitemporal**: Suitable for abrupt changes.
- **Multitemporal**: Captures long-term trends.
- Increasing data availability enhances capability.